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The ROCK

SPRING 2011



**Women scientists
take command**



FROM 'big red' TO BANNER



Technology changes
how University operates

A long time ago (1963 to be exact), in a lab that no longer exists, a young Francis (Fran) Hensler logs on as part of the Slippery Rock University computing staff.

Hensler, with a freshly-minted degree from Florence Utt Business Schools and a tenuous grasp what a computer operator does, has been hired to supervise the University's new IBM punch card tabulating equipment and a data processing unit in the conference room of Old Main.

Just 21 and dressed in button-down corporate, one of Hensler's initial tasks is to relocate the punch card machines to Morrow Field House for SRU's first class registration using rudimentary computer equipment.

"There were only 1,599 students then," Hensler, SRU's long-time software support specialist, said. "All the professors would sit around with punch cards, and there was a card for each seat for each class. A student would go to the professor and say, 'I would like a card for English 101.' The professor would give the student a card. If there were no cards, the class was full.

"It took three days to register everyone," he said.

Higher education has changed in many ways since John F. Kennedy occupied the White House, especially in the application and processing power of computers. From behemoths that occupied an entire room and needed two minutes to process a registration to supercomputers that process 1,000 trillion calculations per second, it has been a remarkable trajectory, one that Hensler

"One of the best features (of Banner) is it is available 24-7. Students drop and add courses all night," Miller said. "Now each student in the networking class in Advanced Technology and Science Hall has three computers in use for the class – a personal computer, a network server and another computer which may be an Internet server."

Miller, one of SRU's who's who of information engineering, joined the University in 1969 – the same year Apollo 11 touched down on the moon. Miller has always been a computer enthusiast, a true believer when few others were eager to jump on the technology train.

"When I started working here in 1969, I knew computers would change how everything was done," Miller said.

Hensler said the power surge has been startling. "A cell phone has much more power than we had in our first computer. It's been a learning process all these years. I would say 50 percent of my time was studying new things my whole career."

"Big red"

The era of mainframes dawned with the Age of Aquarius, in 1968. SRU purchased an IBM 360, a red computer that took up an entire classroom in Spotts World Culture Building. No students used it; booting up in those days meant lacing up football cleats or ice skates.

Larger than a VW Beetle, the 360 paled in processing power in comparison to today's units. Hensler said the 360 possessed 32K



and his colleague Carl Miller, director of administrative information and technology, have helped the University to navigate.

Hensler and Miller helped SRU achieve a computer milestone this spring with the successful launch of the University's \$4.6-million student-information system –the Banner Unified Digital Campus. Banner is the world's most widely used collegiate administrative suite of student, financial aid and finance systems. It is a tightly integrated suite of scalable, enterprisewide applications on a single database. The system facilitates online registration and generates instant reports for academic records, enrollment services and financial aid. It helps the University of more than 8,800 students anticipate trends and improve student learning. It helps students by providing consistent and integrated systems and data.

More than 135,000 student records and 3 million archives had to be migrated from the old legacy system to Banner.



of memory, equaling 32,767 bytes (characters). Today's computers calculate memory in megabytes (1 million bytes) and gigabytes (1 billion bytes). A typical personal computer today contains 2 gigabytes of memory, Hensler said. The new Banner system harbors 430 gigabytes of memory.

SRU used the IBM 360 for registration, student billing, grade reporting and transcripts, although students would find programming cumbersome by today's standards.

"Back then we wrote our programs in IBM cards," Miller said. "When you wanted to write a program, you wrote it on a coding sheet. And then the key punch operator would key punch it in the cards. You'd take a deck of cards and run the cards through the computer, and that's how you wrote the program."

Some students took advantage of the punch card registration system, which involved obtaining a punch card from a professor and then handing the card to the registrar.

"Of course some students did not like some of the instructors so after the student walked away from the instructor, they would just drop that class card on the floor," Miller said. "Students that showed up late would pick up cards from the floor and build a schedule."

Online registration

In 1971, with the counterculture movement in full swing, registrar Joseph Marks issued a difficult challenge: Introduce online student registration by fall 1972. With a handful of new online terminals on campus, Marks assigned the design task to Hensler and Emil Maurer, who wrote the program in basic assembler language (machine code). At that time, all online programs had to be written in BAL and punched into cards – a slow and tedious process.

"We only had six terminals on campus, so all six terminals were moved down to the Student Union," Hensler said. "The terminals in the Student Union were connected by a single telephone wire to the mainframe computer."

The first online registration nearly flopped. "The big problem with the 1972 registration was that we made a bad estimate of how many students we could register per hour. When 9 a.m. came, most of the 8 a.m. students were still being registered," Hensler said. "And it just got worse through the day. I don't know what time it was when the last student was registered, probably after 8 p.m. Carl worked until 3 a.m. trying to speed up R012 (the mainframe) and he was back on the job at 8 a.m. The terminal operators really had it tough. The students were pressing closer and closer to them and I remember trying to push them back."

Still, the program Hensler designed in 1972, R012, was used for the University's summer 2011 registration, although many additions have been made to it. It was adapted for telephone use (RockTalk) and later for online registration via the Internet.

"It is a great testimony to the quality of our programming that many programs were used for more than 40 years, right up until the cut over to Banner."

– Fran Hensler

were used for more than 40 years, right up until the cut over to Banner," Hensler said.

"The program is older than many of you reading this article," Miller said.

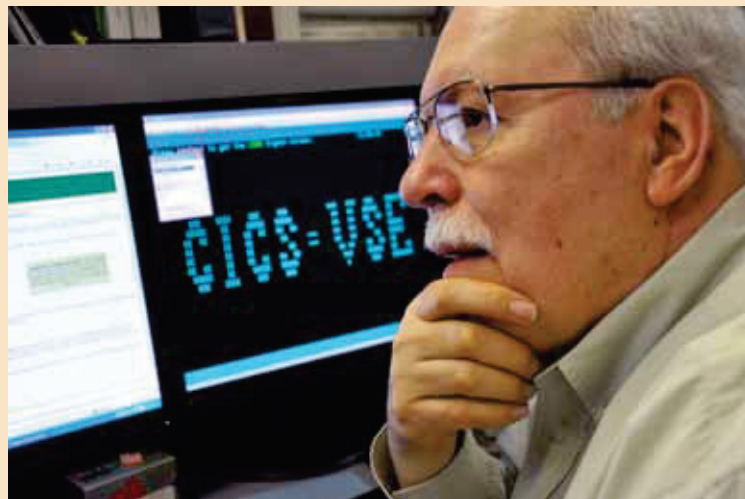
Like anyone who has stayed with the same employer for many years, Hensler and Miller have their share of "stories." Both recalled a near-emergency computer shutdown in the '70s that was caused by someone outside of programming.

"One afternoon all the terminals stopped working, and we couldn't find the problem in the University Union so we headed for Maltby to see if there was a problem with the mainframe," Hensler said. "As we approached Maltby, we saw an open man hole. We were curious about what was going on down there. One of us – almost jokingly – asked if the man had cut any wire and the answer was yes."

They reconnected the wires. "In about an hour everything was back to normal, but we were way behind schedule with many students anxiously waiting to be registered," Hensler said.

Back when Hensler and Miller started working at SRU, only a few SRU students took computer courses. They studied FORTRAN, the first high-level programming language that allowed for the easy translation of math formulas into code.

By the mid 1970s, personal computers were in production,



Fran Hensler

although most college students still used typewriters for writing assignments. Bill Gates introduced the Altair Computer in 1975. Microsoft began marketing the computer in kit form a year later, selling it by mail order through advertisements in Popular Electronics and other hobbyist magazines. Retail stores began selling PCs in 1977, while IBM entered the fray in 1982 by introducing a PC.

SRU enters the world of PCs

Computers took off in the 1980s and especially 1990s, nationally and at SRU. Atari registered the first two computer games, "Asteroids" and "Lunar Lander" in 1980. The WordPerfect 1.0 word processing program came out in 1982. Although computers were still largely insider's technology, Time Magazine named the computer its "Man of the Year" in 1982. Apple introduced the Mac Plus in 1986, selling it for \$2,600.

SRU, ahead of the curve, bought four personal computers in 1980, setting them up in the president's office and vice president's office.

"There was a big learning curve about how to use them," Miller recalled. "The only two pieces of software were spreadsheets and word processing. I remember talking to the ladies over in the vice president's office because it was a priority office. I knew a little wee bit about word processing so I showed them how to type

something in and print it out on a piece of paper. They said they were never going to learn how to do that. A couple weeks later I went back and they weren't having any problems."

Miller said the backspace key on the personal computer was the "greatest thing ever invented." Before backspace, users had to retype a whole page when making corrections. "In a few years all faculty and most clerical staff had a personal computer on their desk connected to the mainframe," he said.

As the University more fully realized computers' potential for student learning, student computer labs opened. The Rocket newspaper switched from the cut-and-paste design method to computer pagination in fall 1988. Computer science became a popular major.

Peek into residence hall rooms, homes or offices today and you will see users drowning in a sea of e-mails, tweets or Facebook posts, a social phenomenon that grew out of the Internet. When



Carl Miller

SRU introduced e-mail in the early 1990s and connected to the Internet in 1991, students increased their computer use so much it nearly crashed the system.

"It just about killed the mainframe because e-mail was so popular with students," Hensler said. "It overloaded the system, so that was one of the first things that was off loaded onto servers."

Miller said it was possible by the mid 1990s to connect phone lines to the mainframe and program a system so it could automatically enter information from a phone keypad to a computer terminal screen. "We added this to student registration and called in RockTalk," he said.

In 2000, SRU had learned enough about the Internet to connect it to the system registration. "The Internet version of RockTalk was faster to use since you could see the message instead of waiting for the voice to say the message," Miller said.

Banner offers new opportunities

Like other outdated computer programs, RockTalk will disappear down the dot.com memory hole and be turned off by the end of summer. All of the information from the old IBM mainframe will be migrated to Banner by the time the fall term begins, Miller said.

Banner helps address the needs of today's learners, servicing them with systems and information that is relevant to them. This allows faculty to better assist students by providing increased automation and efficiency. Further, Banner deploys a set of tools that enable the University to better focus its resources on what matters most by enabling better decision making in the areas of enrollment management, student success and retention.

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As Pennsylvania demographics change, Banner helps Slippery Rock University quickly see trends and adjust to meet the ever-changing needs of students.

Banner offers many practical advantages for students. The system operates 24 hours a day, unlike the legacy system that functioned 8 a.m. to 8 p.m. weekdays. Banner allows class searches by attribute type. If a student wants a night class to fulfill a liberal studies block, he or she searches under that designation.

And Banner more than cuts registration time in half. Banner allows real-sync between Banner and the University's D2L course management system. That means enrollments and course creation and student information that changes in Banner will happen near real-time in D2L. It takes 24 hours with the legacy system.

Students who used Banner for fall class registration, say Banner processes registrations much faster.

"The major benefit I found in the new system is the speed of the entire process," said Chris Kite, a therapeutic recreation major from West Middlesex. "It usually would take me 45 minutes to register under the old system. Using Banner for the first time, I completed registration in less than 10 minutes easily. The system works by checking the classes meeting the requirements designated by the student."

Another advantage to Banner is 1,100 institutions worldwide use it. The old system was customized to the hilt, and was reaching the end of its technological life cycle. SRU is now part of a global network of users.

Financial aid, like other departments, finds Banner a useful tool.

"The new administrative computing system makes a huge impact on our financial aid operations on campus," said Patty Hladio, director. "Our students now have access to the status of their financial aid application and awards all day, every day, which means they can access their information at times most convenient to them and to their families. In addition, we are able to automate more of the required processing functions related to aid administration which will then give us greater opportunities to provide one-on-one counseling and financial aid literacy programs to our students."

Elliott Baker, executive director of academic records and summer school, said he expects Banner to process between 10 and 100 transactions per student by the time the fall semester begins. "When all is said and done, it will have processed between 70,000 and 80,000 individual transactions," he said.

Without a single punch card littering the floor.